Rewind & Amplify

Predicting how hidden defects change quantum measurement patterns

A Qiskit model tested on IBM hardware.
Exploratory extension: laser diagnostics.

Qubits / Florida Gulf Coast University

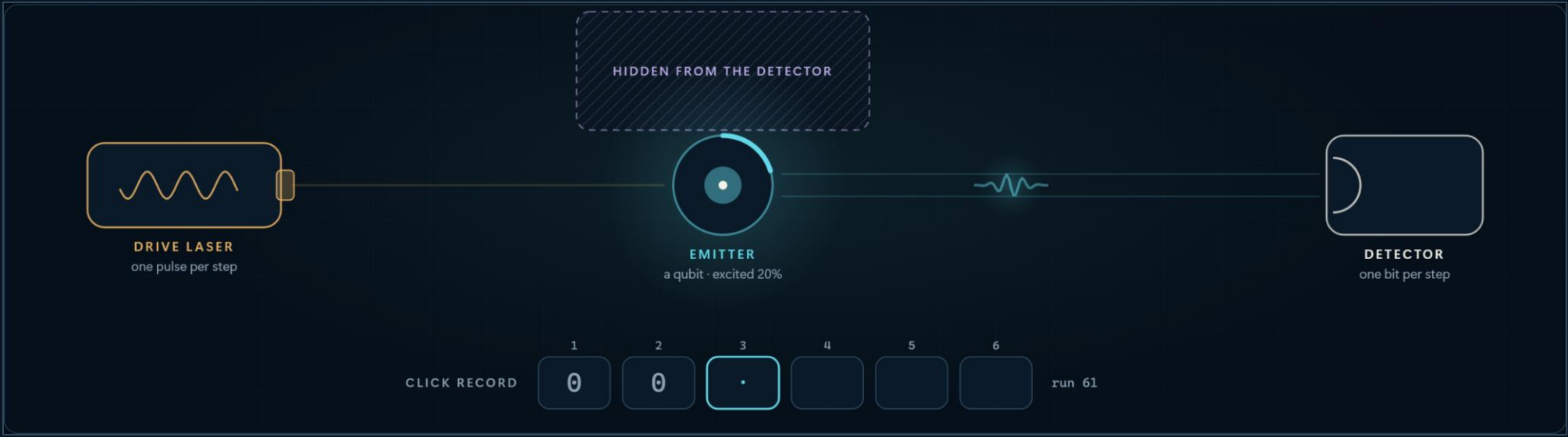
Florida Qiskit Hackathon finals / FAU / 14 November 2026



THE PROBLEM

Something is hidden inside the signal.

A detector records one bit per time step: click, or no click. Out of sight, a defect can trade energy with the source and quietly change the pattern.



Quantum-chip teams

Unseen defects drain their qubits between calibrations.

Photon-source engineers

Two sources can look equally bright and behave differently.

Anyone reading a detector

A detector fault can look exactly like a defect.

Guess the cause wrong, and you fix the wrong thing.

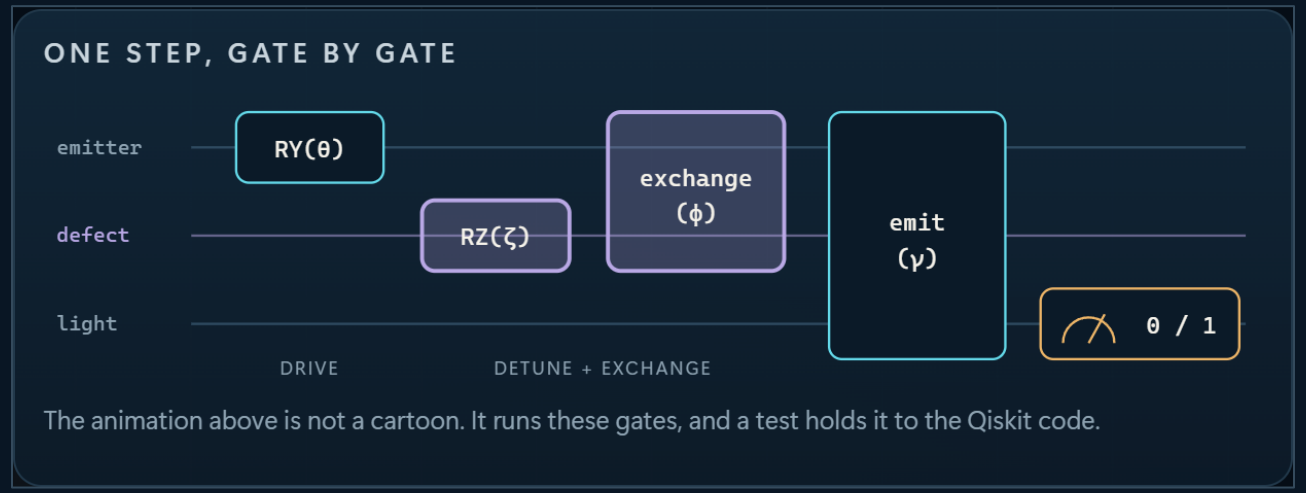
Picture of our model, captured from the live Lab. On IBM hardware, superconducting qubits play these roles; no optical photons were detected.

OUR MODEL

A recurrent machine whose memory is quantum.

We call it a QRSM: a quantum recurrent state machine. Every time step repeats the same four gates.





The animation above is not a cartoon. It runs these gates, and a test holds it to the Qiskit code.

2 qubits

of memory: the emitter and a hidden defect. Never measured.

1 bit

comes out per step: click, or no click.

Each bit

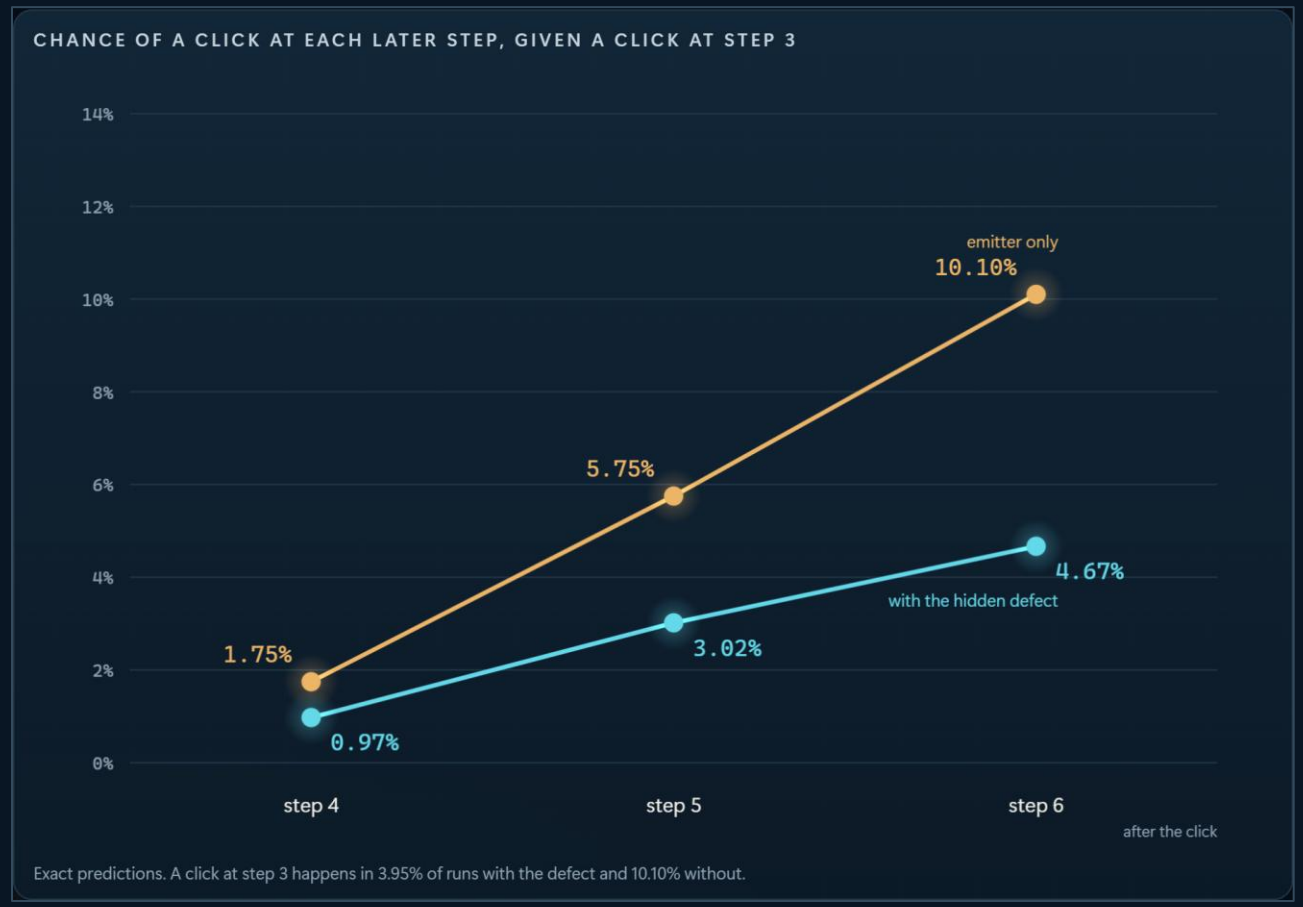
updates the hidden state, so the model predicts what comes next.

Same gates as qrsm_grover/model.py. The picture is a live simulation of that circuit, tested against Qiskit to 10^{-12}.

THE QUESTION

One click. Two futures.

Given a click at step 3, what happens next? With no defect, a click empties the source and the machine starts over. A hidden defect remembers.



EMITTER ONLY

10.10%

chance of a click at step 6

WITH THE HIDDEN DEFECT

4.67%

chance of a click at step 6

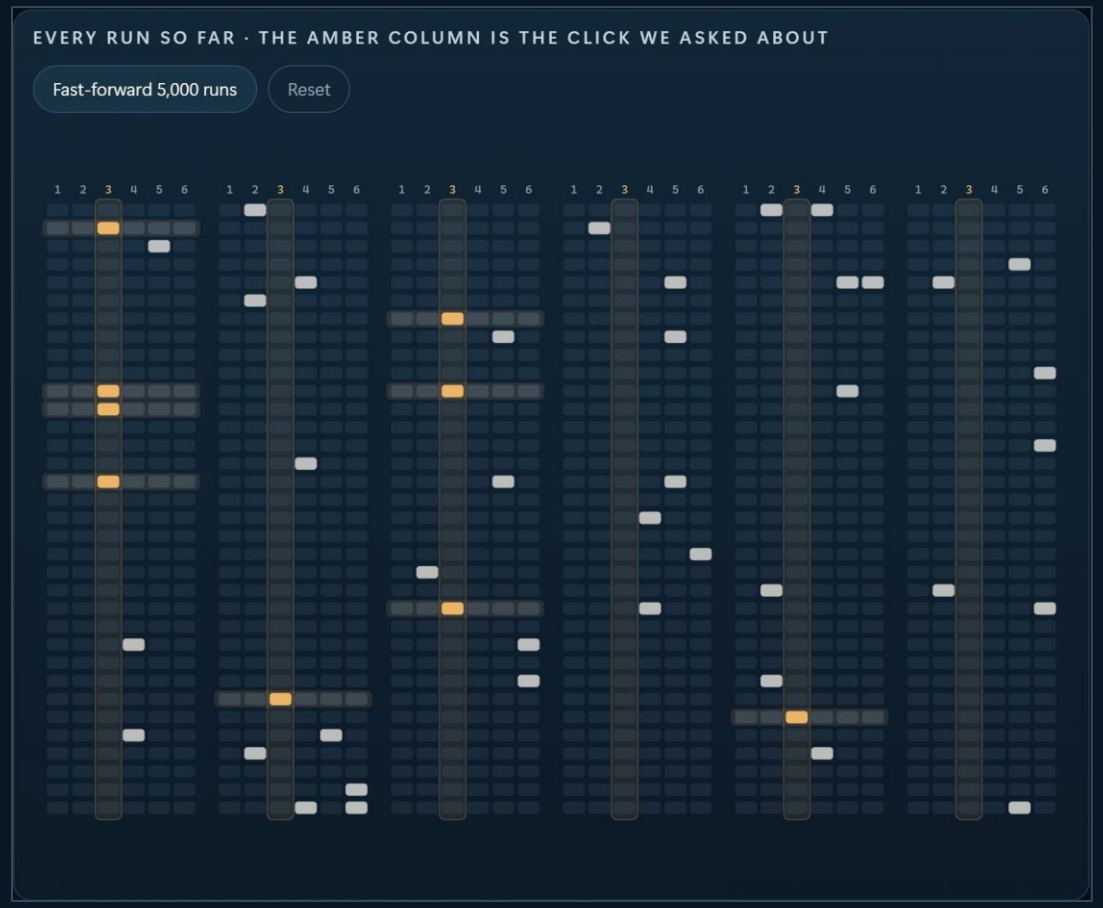
The pattern after a rare click is a fingerprint of hidden memory.

Exact predictions of two models we programmed, recomputed live by the Lab and by reproduce.py. Not a diagnosis of a real defect.

THE COST

The useful runs are the rare ones.

To study what follows a rare click the ordinary way, you run the machine again and again and throw away every run that missed.



3.95%

of runs contain the click we asked about

25 runs

per useful sample. The other 24 are thrown away.

27,495,891 runs

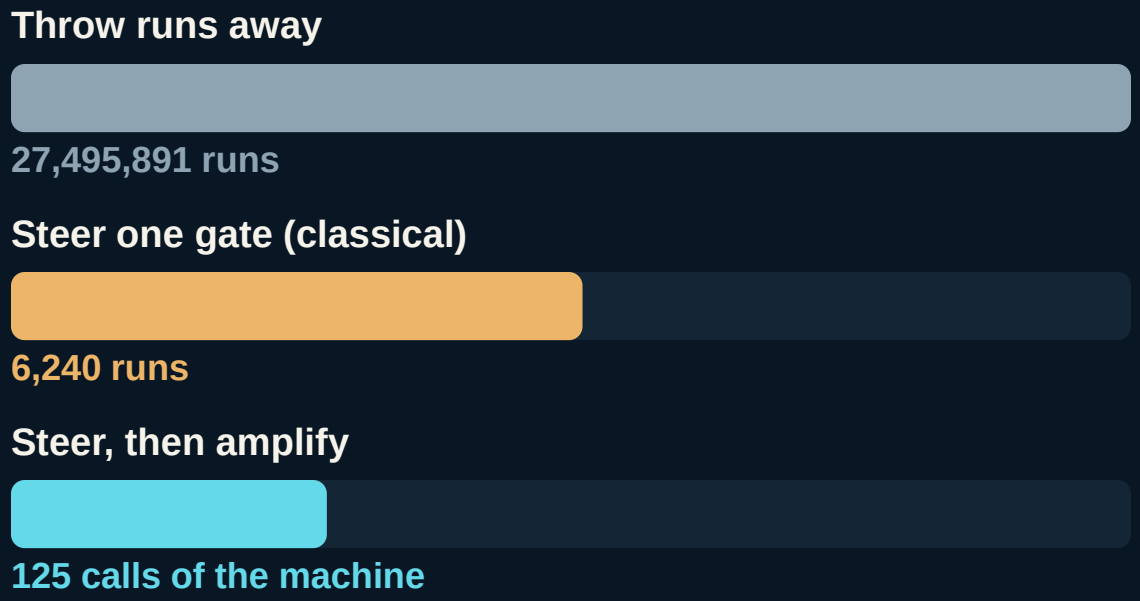
to match one real 64-step record this way

Rejection sampling costs 1/p runs per kept sample. The 64-step figure is an exact calculation for a real record from our first project.

WHY QUANTUM MIGHT HELP

Grover makes rare answers common. Two catches.

COST OF ONE MATCHING SAMPLE · 64-STEP RECORD · LOG SCALE



Rarity costs 1/p the ordinary way, and about 1/√p with amplification.

1 **It must run the machine backwards.**

Our model resets a qubit every step, and a reset cannot be undone.

2 **It is deep.**

On today's hardware every extra gate is a chance for noise.

No speedup today. At this size an exact classical sampler answers in a fraction of a millisecond. Its cost grew about 6× per added hidden qubit in our tests: that is where a quantum method could matter.

Amplitude amplification is prior work (Brassard et al. 2002; Low, Yoder and Chuang 2014). Bars are exact calculations for one 64-step record, not hardware runs.

OUR METHOD

You can’t amplify what you can’t rewind.

1 Purify

A fresh qubit replaces every reset. The whole run becomes one reversible circuit.

2 Steer

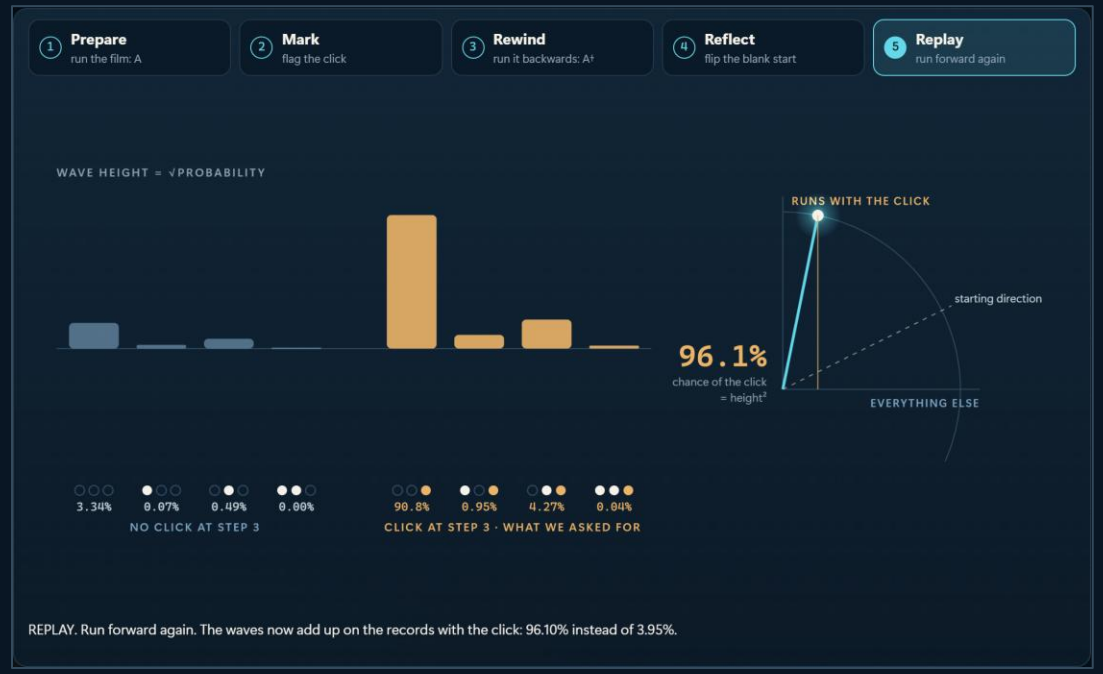
Tune one gate so the rare click is less rare. The answer is unchanged.

3 Amplify locally

One round. The reflection checks 3 qubits, not all 5: about half the gates.

4 Verify

The click is measured directly, so a failed shot shows itself and is dropped.



IDEAL SIMULATOR · ONE ROUND

3.95% → 96.1%

chance of the click we asked about

0.000

error in the answer with the full reflection

0.035

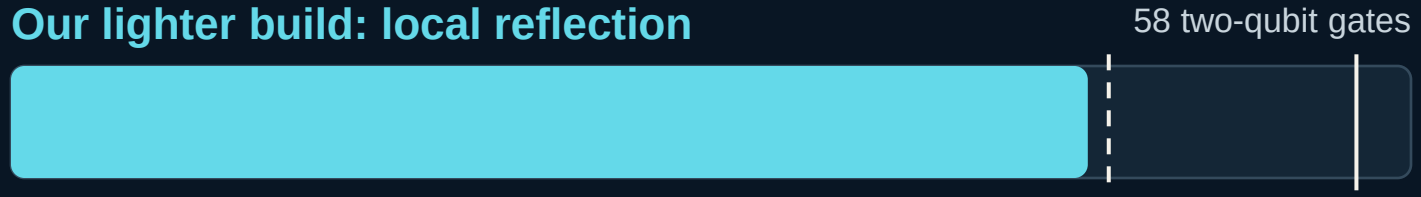
with the lighter local reflection: a small, known bias

What is ours: the mapping onto a hidden-memory model, and the noise-aware build. Qiskit circuits, Aer verification, transpiled to IBM’s CZ basis. 114 tests on the core package.

REAL QUANTUM HARDWARE · IBM MARRAKESH · 11 NOVEMBER 2026

On IBM hardware, fewer gates won.

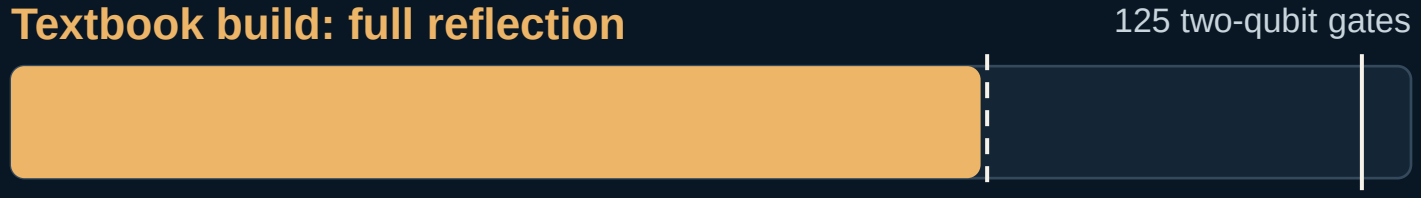
Our lighter build: local reflection



76.90%

6,300 of 8,192 shots hit the click

Textbook build: full reflection



69.26%

5,674 of 8,192 shots hit the click

No amplification: plain sampling



6.74%

solid tick: ideal simulator dashed tick: our noise model (next slide)

+7.6 points

for the lighter build (95% interval 6.3 to 9.0)

0.12 vs 0.18

error in the answer. Lower is better.

7% → 77%

amplification at work on a real chip

8,192 shots per circuit, raw counts, no error mitigation. Job dbqb1ifv5hds73dfo310. Width and routing also differ, so gate count is not proven to be the only cause.

SIMULATOR AGAINST HARDWARE

A simple noise model predicts the gap.

The same compiled circuits that ran on the chip, simulated offline with the chip’s own error rates.

CHANCE OF THE CLICK	Local · 58	Full · 125
Ideal simulator no noise: the two builds tie	96.1%	96.5%
Noise model, calibration only the chip’s reported error rates	83.2%	76.3%
Noise model, echo-fitted gate errors × 1.54, fitted on echo circuits only	78.4%	69.7%
IBM Marrakesh, measured 8,192 shots each	76.9%	69.3%

1.5× noisier

than its calibration said, measured by running the machine forwards and then backwards.

Suppression did not help

Dynamical decoupling and twirling moved the answer error from 0.11 to 0.15 (local) and 0.20 to 0.24 (full).

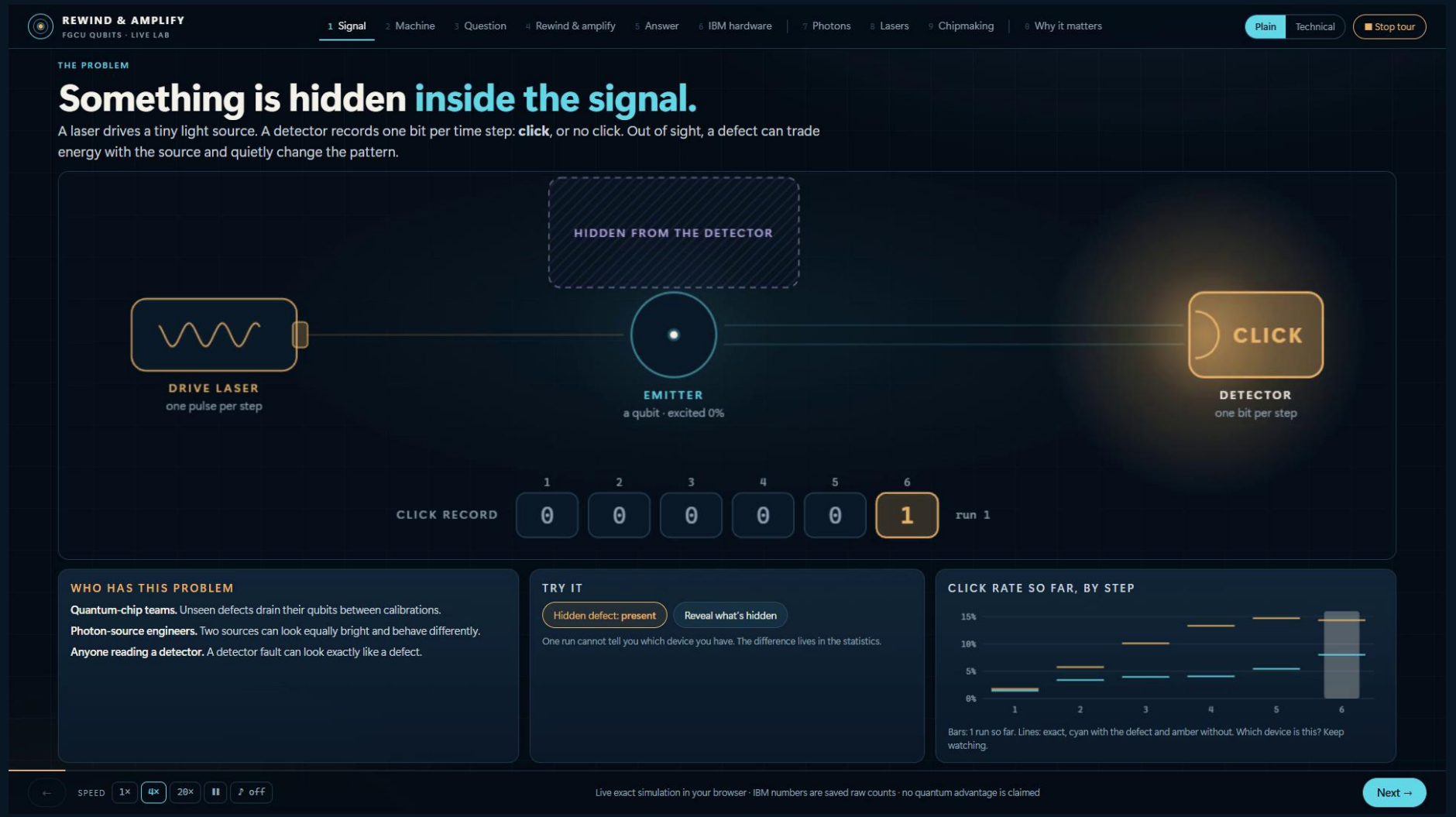
The bias is predicted

After-click answer, local build: 8.8% and 11.7% measured, 8.4% and 11.3% predicted. Exact: 4.7% and 10.1%.

Depolarizing gate errors from the saved calibration, readout errors measured in the same job. Not modelled: idle decoherence, crosstalk, drift.
hardware_campaign/noise_model/

SEE IT RUN

The Lab: our model, live in a browser.



TRY IT

Scan for the code, the Lab and every result.

- Live, exact simulation
- Real IBM counts
- 33 tests hold it to Qiskit and to the saved data

A screen recording of lab/index.html: the model, the method, the IBM result, then three follow-up studies. Open Rewind_Amplify_Lab.html to drive it yourself: one file, no install, works offline.

THE SAME QUESTION, ASKED OF OTHER LIGHT

Beyond qubits: three tests. Honestly scored.

SIMULATED · PUBLISHED DATA REANALYSED



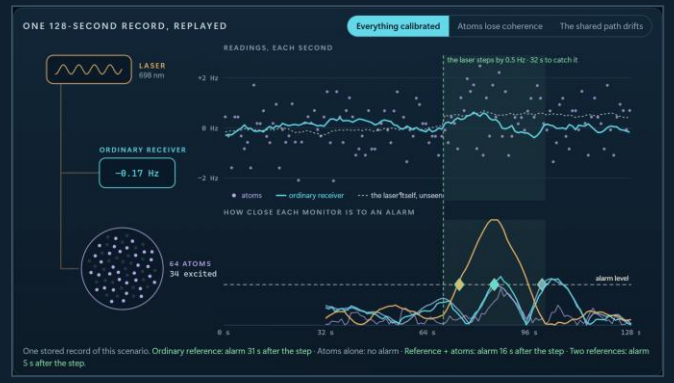
Photon sources

Two sources, equally bright. Only the pairs show which one has a hidden memory.

0.0% vs 0.8%

of attempts split the pair, at identical brightness

SIMULATED · NEGATIVE RESULT



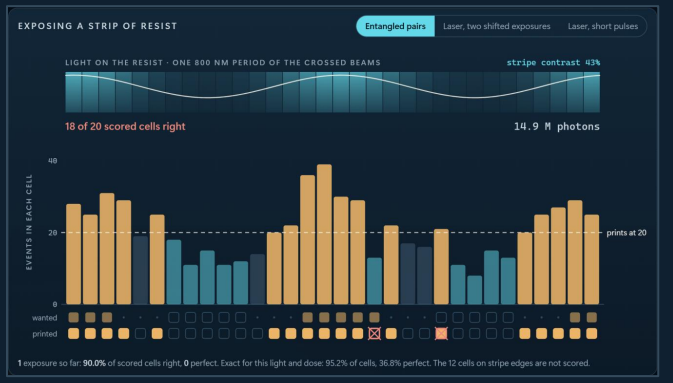
Laser monitoring

Adding an atomic sensor helped a little. A second ordinary reference helped far more.

58% vs 98%

of frequency steps caught within 32 seconds

SIMULATED · AN ASSUMED RESIST



Chipmaking

Entangled pairs print cleaner stripes, and use more light to do it.

95% vs 83%

of cells right, for 18 times the photons

All three are our simulations, plus one reanalysis of published data. No photons, lasers or wafers were measured by our team. Pictures are the Lab's three research tabs.

WHAT WE CAN DEFEND

What we claim, and what we don’t.

WE CLAIM

- A recurrent quantum model made reversible, with identical click statistics.
- Amplitude amplification on its histories, with the answer preserved in the ideal case.
- A lighter, noise-aware build that beat the textbook build on real IBM hardware.
- Every number traceable to saved counts, compiled circuits and tests.

WE DO NOT CLAIM

- A quantum speedup. At this size an exact classical sampler is faster.
- A detected defect. IBM qubits played the roles of a model we programmed.
- Any measurement of photons, lasers or wafers.
- That amplitude amplification is ours. It is established prior work.

NEXT EXPERIMENT

Fit candidate models to held-out, calibrated device records and compare with a strong classical hidden-Markov baseline. Then find where a larger hidden memory makes the classical sampler the bottleneck.

TEAM QUBITS · FLORIDA GULF COAST UNIVERSITY

What we built, and what we learned.

Nicholas Taylor
QRSM research, the Grover fusion, experiments

Timofey Yudintsev
Research, and the keyword-search idea

Dan Izquierdo
Experimentation, and the 45-second video

Dillon Satria Sitepu
Formatting and revising

WHAT WE LEARNED

More hits is not more truth.
A noisy circuit can return the right bit for the wrong reason. Check the answer, not only the hit rate.

On today’s hardware, lighter wins.
58 gates beat 125, in the run and in the noise model.

Report what goes against you.
A conventional sensor beat our quantum add-on, 98% to 58%. It is in the deck.

Code, Lab and evidence

Thank you. Questions?

The QRSM idea and its exact filter predate the event. Amplitude amplification is prior work. Built with AI assistance (Claude, Codex); the team set the questions and checked the results.

REFERENCE

Backups, by the question asked.

ON IBM HARDWARE

15 What each move bought
16 Three runs, one leader
17 Noise model, circuit by circuit
18 Deeper circuits that failed
19 What ran, and where it stops

METHOD AND CREDIT

20 One step, in maths
21 Rerun it: one command
22 Rubric map
23 Ours, prior work, AI

LIGHT: PHOTONS AND DETECTORS

24 Equally bright, not equally good
25 A real quantum dot, reanalysed
26 The detector's blind time
27 A QRSM for photon pairs
28 More data against calibration
29 Better model, more qubits

LASERS

30 Monitors under stress
31 What the monitor assumes
32 Atomic channel on IBM
33 Drive amplitude: negative
34 Where quantum enters

CHIPMAKING

35 Sharper lines, more photons
36 The 389-qubit question
37 Reading a phase
38 Real lithography light

VIDEO

39 The cinematic cut

In a slide show, type a slide number and press Enter to jump to it.

RAN ON IBM HARDWARE

What each move bought, measured on the chip.

First run on IBM Marrakesh, 2,048 shots per circuit, raw counts. The chance that a shot contains the click we asked about.

BUILD	CHANCE OF THE CLICK	TWO-QUBIT GATES
Plain sampling	6.7%	22
Steer one gate	24.1%	22
Textbook Grover: 3 rounds, not steered	48.7%	345
Steer + 1 round, full reflection	64.3%	125
Steer + 1 round, local reflection (ours)	77.4%	58

Steering costs no gates and lifts 7% to 24%. Without it, 345 gates stall at 49%.

hardware_campaign/runs/original_plain_v2/analysis.json. One run of each circuit. The last two were repeated at 8,192 shots (next slide). Ideal values: 3.95% plain, 96% for both one-round builds.

RAN ON IBM HARDWARE

Three runs. The lighter build led every time.

The same two compiled circuits, run three times on 11 November 2026. The runs are separate and never pooled.

RUN	CHANCE OF THE CLICK · LOCAL (58 GATES) AND FULL (125)	LEAD	ANSWER ERROR
First run 2,048 shots · raw counts	77.39% 64.26%	+13.1	0.107 / 0.201
With suppression 2,048 shots · dynamical decoupling and twirling	76.46% 61.96%	+14.5	0.145 / 0.244
Precision repeat 8,192 shots · raw counts	76.90% 69.26%	+7.6	0.118 / 0.181

The lead was 7.6 to 14.5 points. Error suppression did not improve the answer.

The 8,192-shot repeat was chosen after seeing the first run: a follow-up, not a preregistered test. Intervals are shot noise only. Routing, width and timing differ between the two circuits.

RAN ON IBM HARDWARE

The same noise model, circuit by circuit.

After-click answer at step 6. Exact: 4.67% with the defect and 10.10% without. The model was fitted only on echo circuits.

BUILD	TWO-QUBIT GATES	PREDICTED	MEASURED	ORDER ON THE CHIP
Plain sampling	88 / 15	12.5% / 12.1%	**13.4% / 10.1%**	**reversed**
Steer + local reflection	98 / 32	8.4% / 11.3%	**8.8% / 11.7%**	**right**
Steer + full reflection	182 / 56	10.4% / 11.4%	**12.9% / 10.9%**	**reversed**
Memory-only control	93 / 23	8.4% / 3.3%	**8.5% / 4.0%**	**wrong by design**

Each pair is: with the defect / emitter only. Plain sampling fails because false clicks swamp a 4% event.

hardware_campaign/noise_model/results.json. One scale factor stands in for everything the model leaves out. Local build, measured difference: −2.87 points (95% shot interval −3.89 to −1.84).

RAN ON IBM HARDWARE

Deeper circuits returned noise. We kept them.

A three-bin optical model compiled for IBM Marrakesh: one amplification round, 512 shots each, beside our two featured circuits.

CIRCUIT	TWO-QUBIT GATES · LOG SCALE		WHAT CAME BACK
Our lighter build		58	76.9% of shots hit the click
Textbook build		125	69.3% of shots hit the click
Optical query, cutoff 0.03 · 15 qubits		26,749	0 of 512 shots accepted
Optical query, cutoff 0.003 · 18 qubits		54,998	0 of 512 shots accepted

Zero hits proves little.

Even a perfect chip expects 0.65 hits in 512 shots: the event has a 0.128% chance.

The distribution is far off.

Measured against target: error 0.44 and 0.38. A better mathematical model did not give a more useful hardware answer.

Error is the total-variation distance of the measured data-register probabilities from the circuit's target, not a state fidelity.
hardware_campaign/runs/optical_dilation_plain_v2/analysis.json

RAN ON IBM HARDWARE

What ran on IBM, and where it stops.

10 jobs on ibm_marrakesh: 207 science circuits and 40 calibration circuits, 688,640 shots, 244 seconds of processor time.

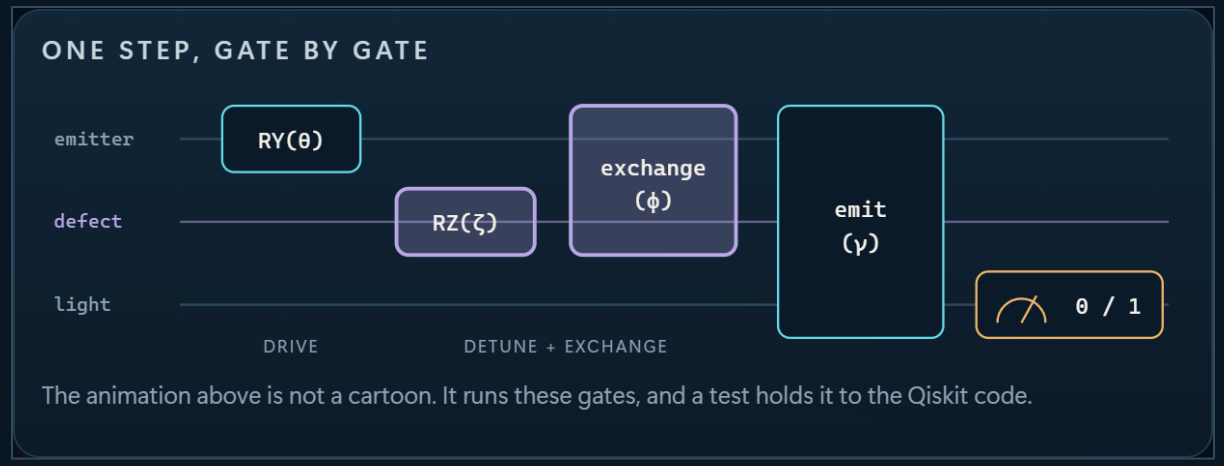
RESEARCH FAMILY	WHAT RAN, OR WAS BUILT	WHERE IT STOPS
The original QRSM query	The full original suite, then an 8,192-shot repeat	The answers are biased, and routing still matters
Quantum-dot, atomic and photon models	Prepared states, channels and joint readout	Digital emulation: no optical setup was involved
Spatial and long queries	Small models, plus deeper stress circuits	A high hit rate can hide a wrong answer
The full 49-cell exposure	A 389-qubit circuit, constructed and checked	**Does not fit: the device has 156 qubits**

All 25 research families have a written ledger of evidence and gaps.

hardware_campaign/results.json, coverage.json and RESULTS.md. Executions include repeats: 207 circuit runs are not 207 findings.

REFERENCE

One step of the machine, in gates and in maths.



ONE STEP	$U = \mathrm{Exchange}(\varphi) \cdot RZ(\zeta) \cdot RY(\theta)$
NO CLICK	$K_0 = (\lvert 0\rangle\langle 0\rvert + \cos\alpha \lvert 1\rangle\langle 1\rvert)\, U$
CLICK	$K_1 = \sin\alpha \lvert 0\rangle\langle 1\rvert\, U, \quad \sin^2\alpha = \gamma$
A RECORD	$P(y_1 \ldots y_T) = \lVert K(y_T) \cdots K(y_1) \lvert 00\rangle \rVert^2$

DYNAMIC VERSION

3 qubits

Measure the light qubit and reset it every step. Simple, but a reset cannot be undone.

PURIFIED VERSION

2 + T qubits

Keep each step's light in a fresh qubit and measure at the end. One reversible circuit, with identical click statistics.

WHAT A CLICK DOES

back-action

K_1 empties the emitter and leaves the defect conditioned on the record. No click also updates the state. That is the recurrent update.

rewindandamplify-main/qrsm_grover/model.py. Steps are dimensionless. The optical labels are an interpretation of the model: the IBM circuits did not measure photon arrivals.

REFERENCE

Anyone can rerun it. One command.

```
$ python reproduce.py

1. The exact model                          with the defect   emitter only
   click at step 6, given one at step 3              4.67%        10.10%

2. The Qiskit circuits on Aer                hit rate   answer error
   plain sampling                               3.95%         0.0000
   steer + full reflection, 1 round            96.49%         0.0000
   steer + local reflection, 1 round           96.10%         0.0353

3. Classical baseline: exact sampler, a fraction of a millisecond per sample

4. IBM Marrakesh, job dbqb1ifv5hds73dfo310      ideal   noise model   measured
   steer + local reflection    58 gates    96.10%     78.40%     76.90%
   steer + full reflection    125 gates    96.49%     69.72%     69.26%

Done in about 10 s. Nothing above claims a quantum speedup.
```

No IBM account, no network. Model numbers are recomputed with Qiskit; measured numbers are recounted from the saved raw counts.

REFERENCE

The rubric, and where each point is answered.

CRITERION	POINTS	WHERE THE TALK ANSWERS IT
A clear problem, and who it affects	**15**	Slides 2, 4 and 11: the hidden defect, who has the problem, two futures
Why quantum, and honest limits	**20**	Slides 5, 6, 7, 9 and 12: the cost of rarity, two catches, no speedup today
Qiskit implementation that runs	**20**	Slides 3, 7, 8 and 10; backups 20 and 21: gates, four moves, one command
Results, interpreted and validated	**10**	Slides 4, 8 and 9; backups 15 to 17: exact, simulator, noise model, hardware
Originality	**10**	Slides 7 and 13; backup 23: the reversible recurrent model and the noise-aware build
Clear to every audience	**15**	13 main slides, one picture each; the Lab's plain and technical captions
Questions, teamwork, learning	**10**	Slide 13; backups 14 to 39: roles, lessons, and a slide for each likely question

Bonus: simulator +1 (slides 7, 9). Real IBM hardware +3 (slide 8). Hardware compared with a simulator, noise analysed, suppression tried +5 (slides 9, 16, 17). Evidence for the judges, not a predicted score.

REFERENCE

What is ours, what is prior work, and where AI helped.

ORIGIN	WHAT BELONGS THERE
Established prior work	Amplitude amplification (Brassard et al. 2002) and amplified conditional inference (Low, Yoder, Chuang 2014)
Before the event	The QRSM idea, the physics-shaped model and its exact filter
The hackathon project	The Qiskit build, the fusion with Grover, steering, the local reflection, and the experiments
Finals work	Model audits, the joint-photon model, the IBM Marrakesh campaign, the noise model and the Lab
AI assistance	Original project and V3: Claude. V2 research extensions: Codex with OpenAI models. The team set the questions and reviewed the results.

A sensing benefit and a computing speedup are separate claims. We make neither today.

Atomic laser-noise spectroscopy also predates this project (Bishof et al. 2013). No new sensing principle is claimed. “Independent checks” means separate implementations inside the project, not external peer review.

SIMULATED BY US

Equally bright. Not equally good.

Two model photon sources send the same number of photons in the same mix of colours. Only the pairs differ.



SOURCE A · ENVIRONMENT HOLDS STILL

0.0% of attempts put one photon in each detector

photon overlap 1.00

SOURCE B · ENVIRONMENT FORGETS

0.8% of attempts put one photon in each detector

photon overlap 0.60

THE SAME TWO CASES, AS CIRCUITS ON IBM

0.81 vs **0.52**

measured overlap · ideal 1.00 and 0.60

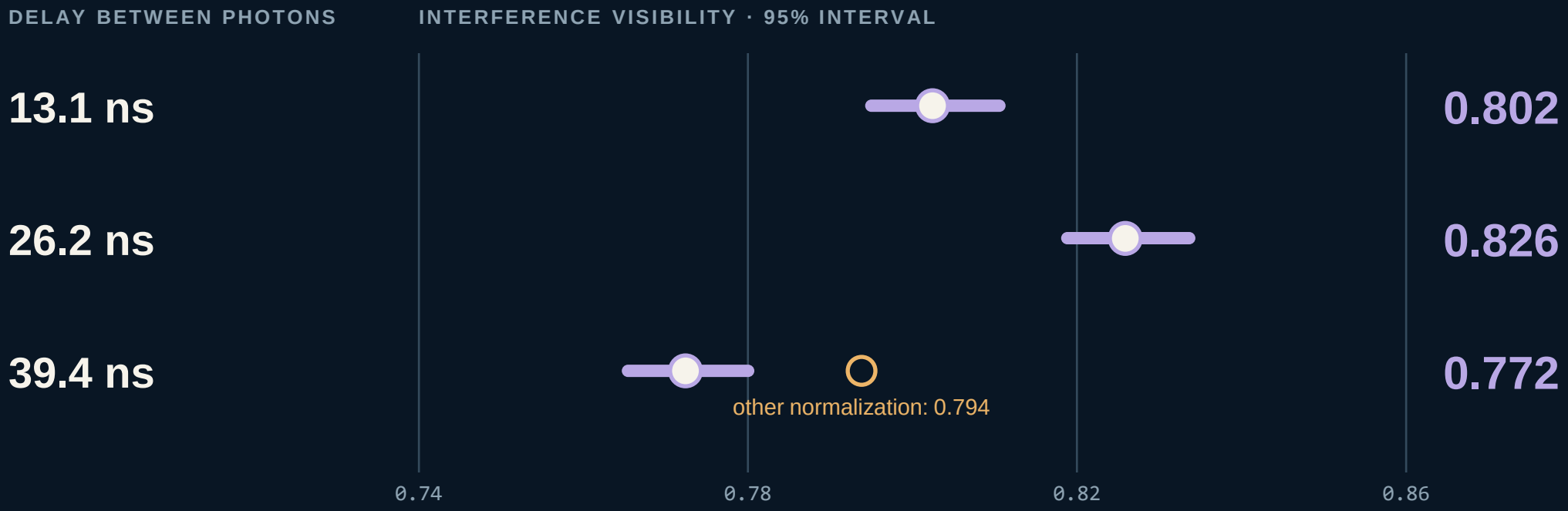
One detector alone sees no difference. The pair pattern does.

A classical two-state environment, 20% photon survival, ideal single photons. No semiconductor defect was identified. quantum_dot_bridge/results.json; picture captured from the Lab.

PUBLISHED DATA, REANALYSED

A real quantum dot: published data, reanalysed.

How well two photons from a GaAs quantum dot interfere, at three delays between them. Measured by Münzberg et al. (2022), CC BY 4.0.



At 39.4 ns, changing the normalization moves the answer by 0.021. The interval is ±0.008.

A source-quality number depends on the measurement controls. That discrepancy says: check polarization, alignment and normalization before naming a cause.

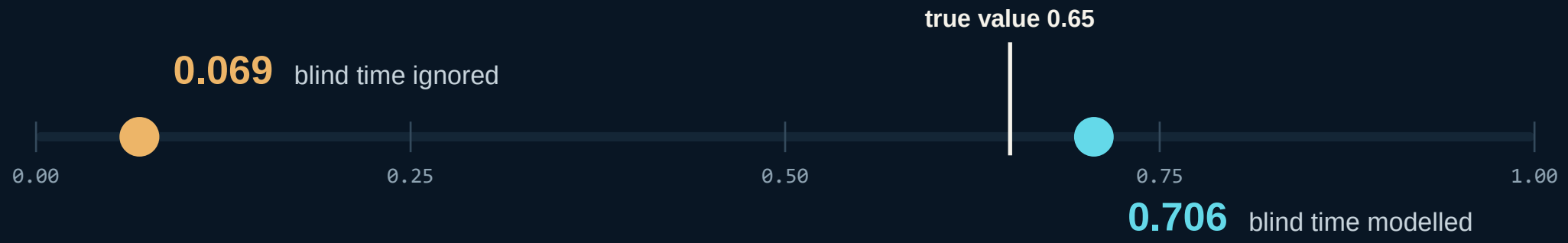
Our peak-area reanalysis of their published histograms: not our measurement, and not a QRSM fit. Intervals cover counting and calibration only.
quantum_dot_bridge/hom_analysis/results.json

SIMULATED BY US

Ignore the detector’s blind time, and the fit is wrong.

Fresh synthetic click records with a known defect coupling of 0.65. The same records, fitted twice.

FITTED DEFECT COUPLING



−0.451

nats per record against a classical hidden-Markov model, with the blind time ignored: clearly worse

+0.008

once the blind time is modelled: level with the classical model, within uncertainty

0.458

nats per record gained just by modelling the detector properly

After a click the detector is blind for one step. Leave that out, and the model blames the defect.

24-bin records. Synthetic inference, not a diagnosis of a physical defect. The corrected fit still pins dephasing at its bound. qrsm_photon_poc/followup/

SIMULATED BY US

A QRSM for photon pairs: memory between pairs.

A model in which the hidden environment persists from one photon pair to the next, so a run of outcomes can carry memory.

1 Emit each photon

Its shape in time depends on the state of the environment.

2 Observe a pair

A controlled-swap parity check gives the beam-splitter statistic.

3 Predict the next pair

The environment is kept, so the record so far shapes what comes next.

SIMULATION, 64 PAIRS PER RECORD

5.6% vs **4.5%**

of records flagged: coherent source, then a classical one. Barely apart.

FOUR PAIRS, AS CIRCUITS ON IBM

0.25 and **0.29**

error of the measured flag distribution. Implementation noise is still large.

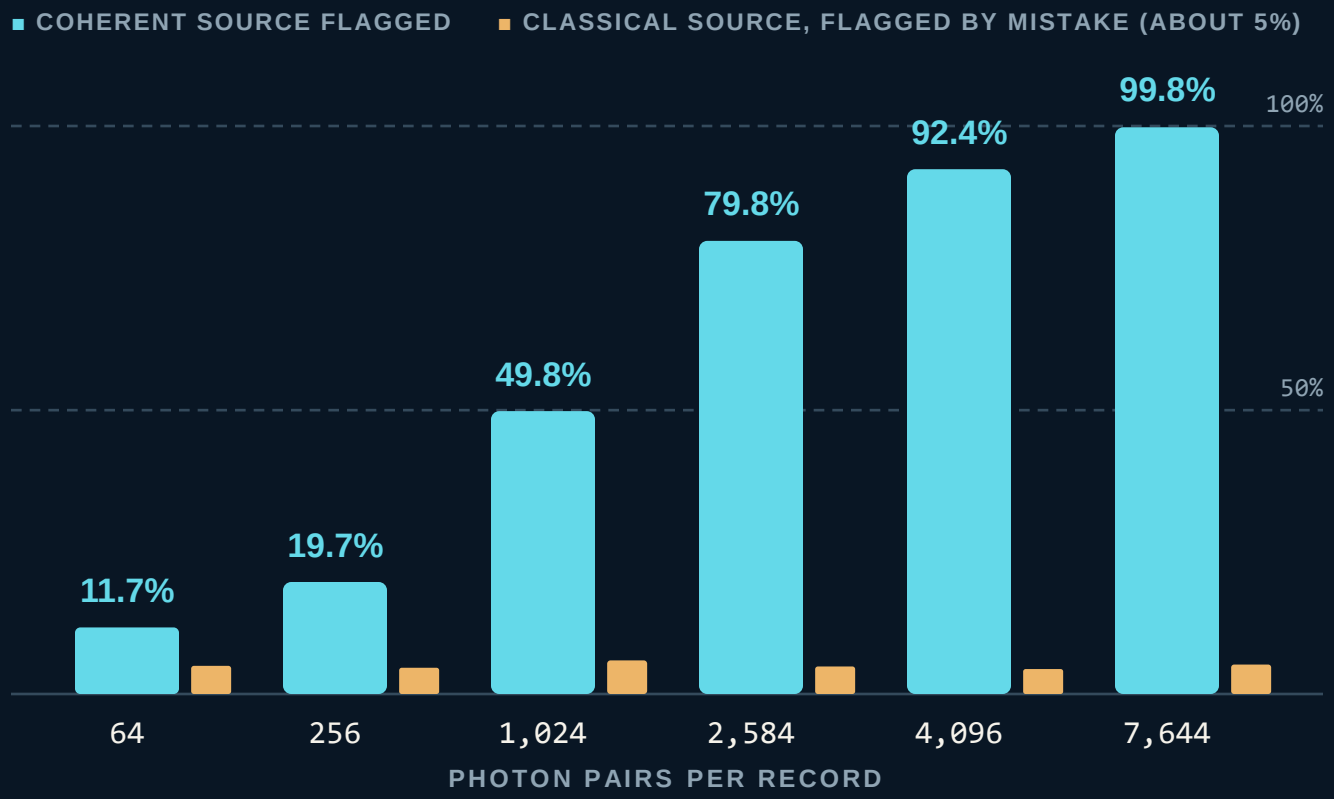
With 64 pairs the test hardly fires. See slide 28 for what more data buys, and what it does not.

quantum_dot_bridge/qrsm_hom_followup/. Flags are a calibrated test for coherent memory. The IBM error is the distance of measured from ideal flag probabilities. Neither identifies a physical defect.

SIMULATED BY US

More data helps only if the calibration holds.

Simulated records flagged as carrying coherent memory, by photon pairs per record (2,000 records each).



But at 7,644 pairs, a classical source is flagged anyway

72% if the collection efficiency changed

53% if the mode overlap changed

21% if the readout changed

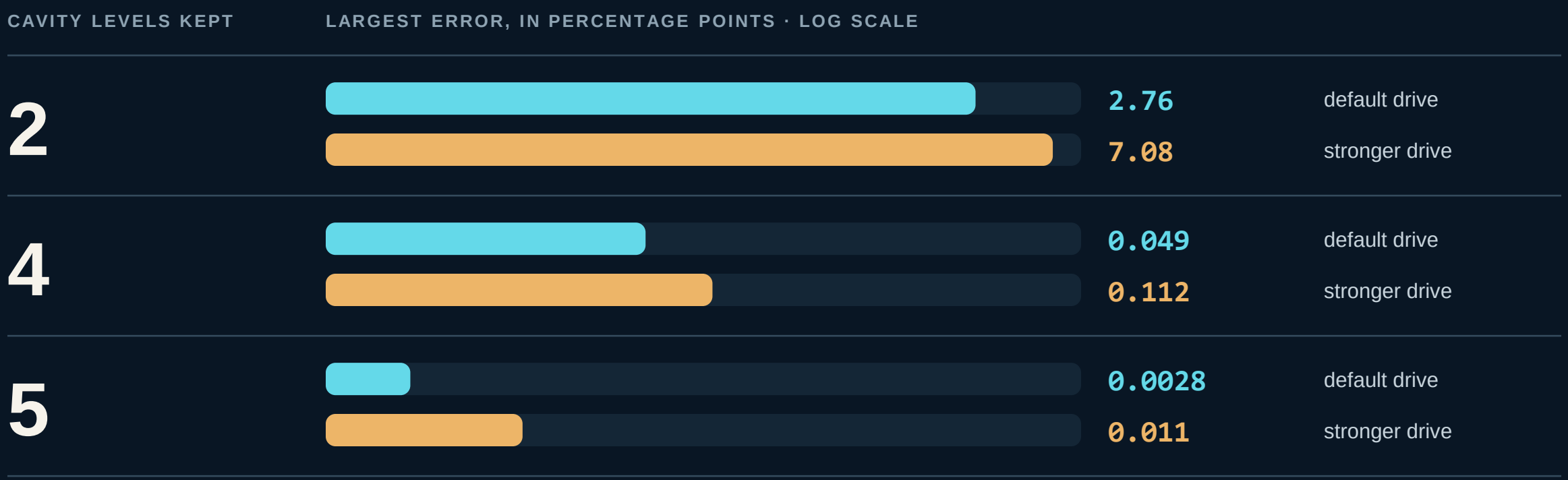
More data cannot repair a wrong model of the measurement.

A selected control and ideal resets between four-pair blocks: a different protocol from the 64-pair study. A stronger 8-state classical model predicts as well within uncertainty. quantum_dot_bridge/qrsm_hom_followup/exploratory_design/

SIMULATED BY US

A better model costs more qubits.

The optical circuit keeps only a few energy levels of its cavity. Largest error in the after-click prediction, against a six-level reference.



Two levels are off by several points. Five are accurate, and need more qubits and gates.

Model accuracy is part of the resource budget: the cheap circuit answers a slightly different question.

A finite truncation study with a six-level reference, not proof of convergence. The circuits that ran on IBM used two levels.
qrsm_photon_poc/followup/dynamics_results.json

NEGATIVE RESULT

A calibration slip can reverse the conclusion.

A laser steps by half a hertz at an unknown time. Drifts caught within 32 seconds, and false alarms on quiet records. 1,920 simulated records per cell.

SITUATION	ORDINARY REFERENCE	REFERENCE + ATOMS	TWO ORDINARY REFERENCES
Everything calibrated	51.4% false alarms 5.4%	58.0% false alarms 5.2%	98.0% false alarms 4.1%
Atoms lose coherence sooner	52.8% false alarms 6.0%	47.3% false alarms 6.1%	98.1% false alarms 5.0%
The first reference steps	44.0% **false alarms 91.5%**	45.6% **false alarms 84.5%**	97.7% false alarms 4.8%
The shared path steps	45.6% **false alarms 91.1%**	48.2% **false alarms 94.1%**	56.1% **false alarms 100.0%**

Calibrated, the atoms add 6.7 points and a second reference adds far more. Degraded, the atoms hurt.

A step in the shared path looks exactly like a step in the laser: every monitor raised the alarm.

Atoms alone caught 7.7%. Calibration is frozen before the stress cases. laser_sensor_viability/followup/main/results.json

REFERENCE

The laser monitor: what we assumed.

MODEL ELEMENT	ASSUMED SETTING
Laser	698 nm, narrow-line: a laboratory wavelength atoms can read
Atomic sensor	64 independent atoms, not entangled
One reading	2.5 ms pulse, 20 ms dark time, 2.5 ms pulse
Record	128 s long, with 975 ms of dead time in every one-second cycle
Atom quality	coherence 200 ms, lifetime 100 s, readout errors 2% and 4%

Time and task are matched across the monitors. Total apparatus is not.

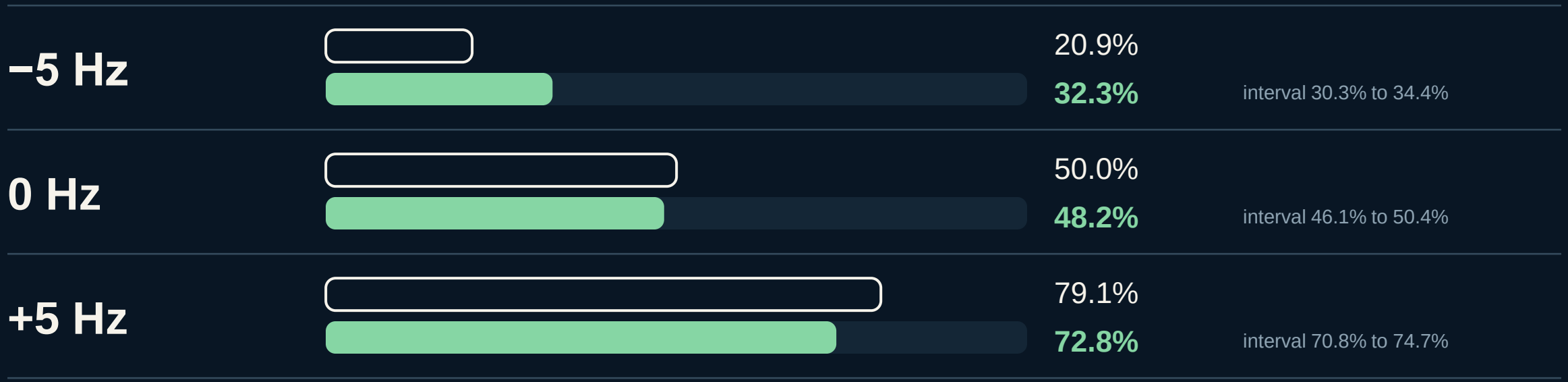
So this is a test of added information, not a resource-matched advantage claim.

Photon allocation is assumed bookkeeping. Reference lasers, cooling, trapping and readout hardware are not costed. Atomic laser-noise spectroscopy is prior work (Bishof et al. 2013). laser_sensor_viability/results.json

RAN ON IBM HARDWARE

The atomic channel, run as a circuit on IBM.

How the model atoms respond when the laser is a few hertz off, compiled to 4 qubits and 22 two-qubit gates. 2,048 shots per setting.



The response goes the right way, and is biased at the edges.

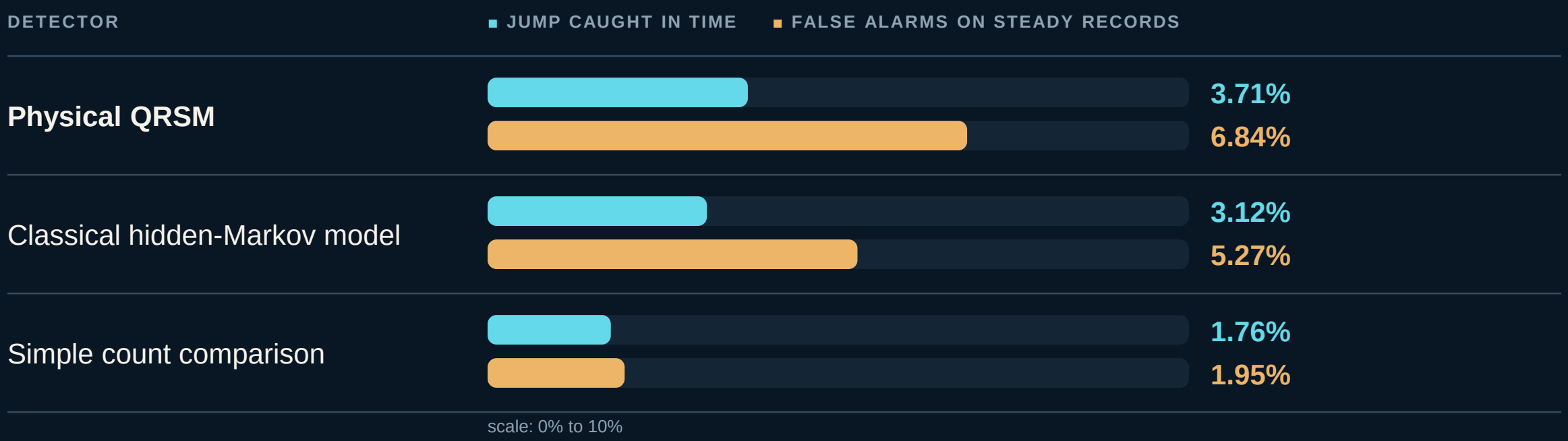
Separately, a two-photon state at phase 0.35 ran as a circuit with an occupation error of 0.033. These are implementation tests of bounded models. Acquisition, calibration and apparatus cost remain optical questions.

Superconducting qubits emulated the model: no atoms, no optical coupling and no 128-second monitor were measured. Raw counts. hardware_campaign/runs/optical_controls_plain_v1/

NEGATIVE RESULT

Drive amplitude: a negative result we keep.

Can a click record reveal a 20% jump in drive strength at an unknown time? Three detectors read the same bits.



ERROR OF THE AMPLITUDE ESTIMATE

0.1136 → 0.1125

ignoring the data, then using it

Every detector raised more false alarms than detections.

Most of the apparent gain came from the prior, not from the clicks.

Field-amplitude proxy on 96-bin records, 512 records per case, one operating point, increases only. The prior-only control was added after the holdout, so it is descriptive. qrsm_photon_poc/drive_diagnostics_followup/

Where the quantum part actually enters.

LAYER 1

Sensor dynamics

How the atoms respond to the laser.

RAN ON IBM AS A CIRCUIT

A Bloch model, and that channel as a circuit on IBM

LAYER 2

Diagnostic inference

Combining the readings into an alarm.

CLASSICAL

Classical, on synthetic records

LAYER 3

Rare-history query

Amplifying a rare condition in a coherent model.

RAN ON IBM

Qiskit circuits, on IBM hardware

Running one component on a quantum computer is not an end-to-end advantage.

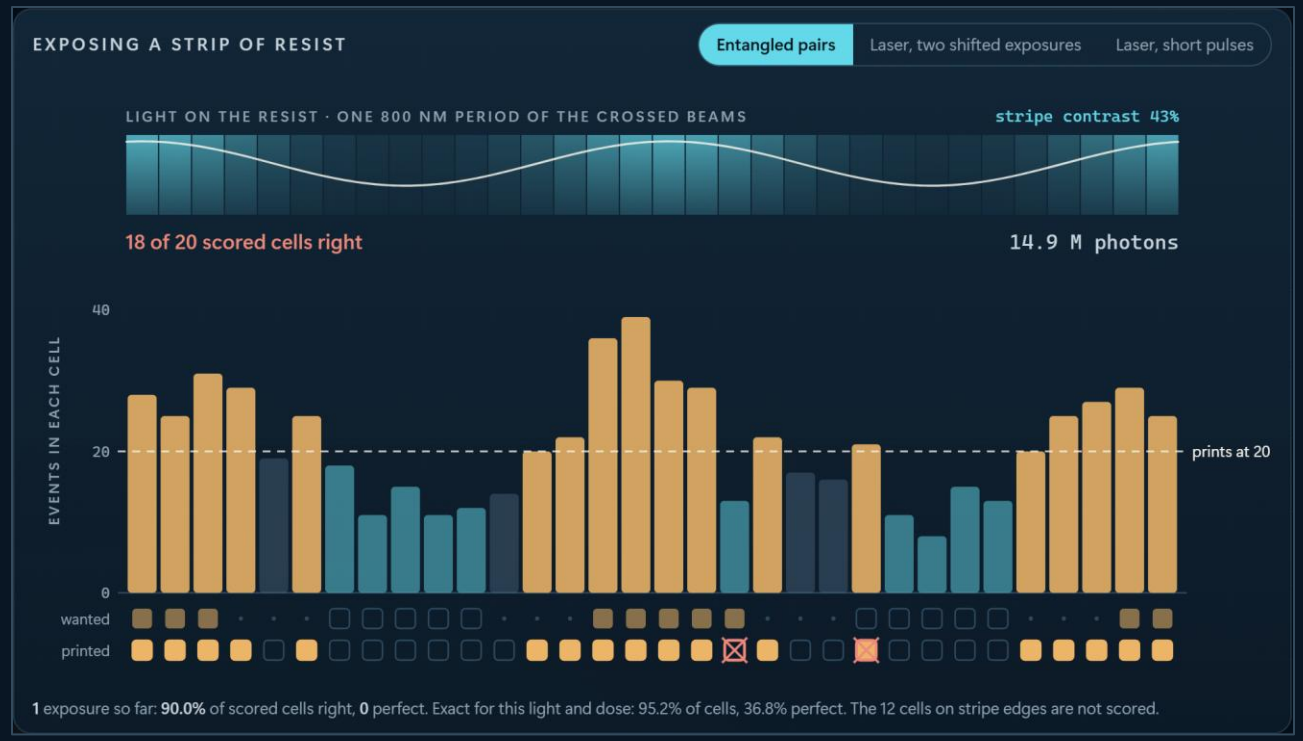
A sensor can use quantum dynamics while a classical computer reads its output. A sensing benefit would not be a computing speedup, and we claim neither.

Individual channels ran as digital circuits. The full acquisition-and-diagnosis benchmark is a separate classical simulation.
hardware_campaign/optical_controls/README.md; LASER_DIAGNOSTICS_ROADMAP.md

SIMULATED BY US

Sharper lines, paid for in photons.

A strip of resist that needs two photons at once, under three kinds of light, each at its own best dose. 20 scored cells.



LIGHT	CELLS RIGHT	ALL 20	PHOTONS
Entangled pairs	**95.2%**	36.8%	14.9 M
Laser, two shifted exposures	**78.3%**	0.7%	6.81 M
Laser, short pulses	**83.3%**	2.5%	0.84 M

The fine stripes appear for laser light too. Pairs give them about twice the contrast.

Pairs print the cleanest stripes, for 18 times the photons of the pulsed laser.

An assumed resist, not a material on the shelf. With 20% transmission, a leakier resist or a 1% pair source, laser light wins.
quantum_lithography_poc/exposure_followup/results.json; picture captured from the Lab.

BUILT AND CHECKED · NOT RUN

The full question needs 389 qubits. We have 156.

Our rare-event question, asked of a 49-cell exposure: how often does a contact fail to print? Built as a reversible circuit, and checked.

QUBITS



- **196** counting photons in each cell
- **23** adding up the score
- **170** spare, for reversible arithmetic



the chip we used: 156 qubits

3.4 million

two-qubit gates, as an upper bound. Our best run on the chip used 58.

0.42%

of exposures: the rare event the circuit would amplify. One ideal round would raise it to about 3.8%.

Solved classically

Dynamic programming already answers this independent-cell model, quickly.

A map of the cost, not a result: building the circuit shows neither an advantage nor that the process works.

Gate counts are constructive upper bounds for one amplification round, not a routed compilation. Never simulated in full or run. finals-lithography/full_spatial_circuit/results.json

SIMULATED BY US

Reading a phase: the benefit depends on the budget.

Measuring one optical phase with 20,000 photons at 81% efficiency. Every photon sent is charged.

LIGHT	ERROR IN THE PHASE, RADIANS · LOG SCALE	WRONG HALF-TURN
Laser light	0.0079	0 of 2,000
Entangled pairs alone	2.23	50.4%
10% laser + 90% pairs	0.0064	0 of 2,000

Pairs alone cannot tell a phase from that phase plus half a turn. A small laser pilot fixes it.

The hybrid is slightly more precise than laser light, and only at high efficiency.

If the pair source succeeds 1% of the time, the same dose takes 100 times longer.

Synthetic task with a known reference, ideal photon counting and declared losses. Zero observed errors is not a guarantee.
quantum_lithography_poc/phase_inspection_followup/results.json

REFERENCE

Lithography light is not our laboratory laser.

LIGHT	ITS ROLE IN MAKING CHIPS	IN OUR MODEL
193 nm excimer laser	The deep-ultraviolet exposure source	**Not represented**
CO_2 drive laser	Heats tin droplets into a plasma	**Not represented**
13.5 nm extreme ultraviolet	The plasma’s light, which exposes the wafer	**Not represented**
698 nm laboratory laser	A narrow line that atoms can read	**A synthetic stand-in**

Conventional source monitors already exist, and already work.

What we tested is whether an atomic sensor adds information, on a stand-in wavelength. Transfer to a real lithography source would need its own experiment.

Roles of the sources: ASML (light and lasers; EUV systems). Source monitoring and wafer effects: Cymer and imec, SPIE Advanced Lithography 2014. LASER_DIAGNOSTICS_ROADMAP.md

REFERENCE

The cinematic cut. 45 seconds.



A cinematic explainer with original music, not a screen recording of the product. The model picture is conceptual; the IBM numbers are the measured ones.